National University of Computer and Emerging Sciences
Lahore Campus

DLD (EE1005)

Date: April 6th 2024

Course Instructor(s)

Mr. Kamran Jamal

Ms. Tamania Javaid

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 45

Total Questions: 3

[redacted]

Attempt all the questions.

CLO #3: Design combinational circuits using functional blocks

Q1: **Design** a car safety alarm considering four inputs (one bit each) [15 marks]

- Door closed (D)
- Key in (K)
- Seat pressure (S)
- Seat belt closed (B)

The alarm (A) should sound if
The key is in and the door is not closed
OR
The door is closed and the key is in and the driver is in the seat and the seat belt is not closed

Construct the truth table and implement the circuit using 8x1 MUX.

CLO #3: Design combinational circuits using functional blocks

Q2: **Design** a combinational circuit that take a 3-bit number as input and generates its 2's complement as output. Implement the circuit using 3-8 line decoder with active low enable and active high output and minimum additional logic. [15 marks]

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CLO #3: Design combinational circuits using functional blocks

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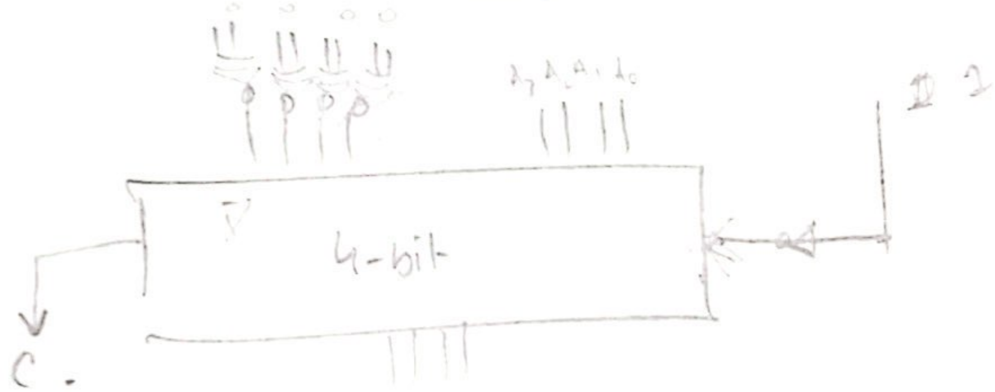
Q3: **Design** a digital circuit to perform the arithmetic operation $x + 3$ if a selection line S is high and $x - 3$ if S is low. Here x is unsigned 4-bit number. Available resources are half adders, full adders and any two input gates. Show complete working. [15 marks]



$B \oplus 0 =$

$X_3 \quad X_2 \quad X_1$

$A_3 \quad A_2 \quad A_1 \quad A_0$

$B_3 \quad B_2 \quad B_1 \quad B_0$